

government has also laid emphasis on the growing need for women safety by installing CCTVs in public areas, buses, metro stations and the like.

The government of India has invested around ₹ 48,000 crores in order to introduce Smart City initiative that aims to create at least 100 smart-cities, where public safety will be given paramount importance by city administrations. This would include protection against crime, natural disasters, accidents and terrorism. Smart City initiative imbibes four major rules of planning: social, physical, institutional and economic infrastructure.

One of the main pillars of institutional infrastructure is public security and safety. Technologically-driven surveillance systems are easy to scale up with a wide range of security systems that work on a single interface. These successfully consolidate physical safety needs ranging from video surveillance, intrusion alarms and access-control techniques, and conserve existing analogue systems. Thus, advanced surveillance techniques that are mostly driven by technology are considered highly cost-effective, as these can be further upgraded and come with low maintenance cost.

There has been a major shift in the demand for surveillance techniques from developed to developing nations in the past decade. Countries gaining from this change are parts of Asia, Eastern Europe, Africa and the Middle East where security markets are relatively untapped. Such an increase in the employment of security services is likely to be sustained by improved economic activities, growing construction activities, growing urbanisation and higher per capita income.

Improved economic productiv-



Fig. 4: Trend in the shift of security market from developed to developing nations (Source: Grant Thornton, 2015)

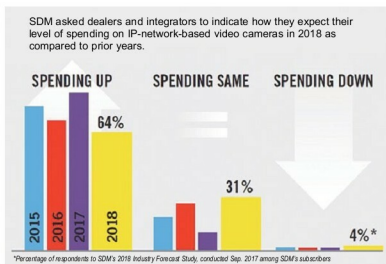


Fig. 5: Expected level of spending on IP-based video cameras in 2018 (Source: www.sdmag.com)

ity followed by rising per capita income promotes employment and installation of modern and advanced technology-driven surveillance systems. Also, manned security dominates demand in developing nations, whereas developed markets are focused more on electronic security services that are more widely used in the US, Canada, much of western Europe, Japan and South Korea. There is an overall shift in market share from western countries to the Asia-Pacific. Between 2002 and 2019, there is a steady increase in Asia-Pacific market and a gradual fall in North America and Europe markets, as can be seen from Fig. 4.

Growth drivers

Electronic security and surveillance have become alarmingly popular owing to the plethora of benefits that these offer. These not only protect against outside break-ins and

burglaries but also play a critical role in sustaining businesses and creating a safer place to live and work. More often than not, investing in a few surveillance technologies can yield a high return on investment for an organisation. Increasing demands for VCA and VSaaS, incorporating artificial intelligence (AI) and video analytics, and amalgamation of electronic security with advanced technologies, such as drones and body cameras, are some of the key opportunities that propel the growth of the market.

According to 'Digital Security and Surveillance' report by Intel, retailers in North America lose up to US\$ 45 billion a year due to employee theft, shoplifting and organised crimes. With the advent of surveillance technology, a huge number of retailers today rely on video and other forms of surveillance techniques to identify suspicious behaviour. Transformation from analogue-based surveillance to IP-based digital technologies has not only made surveillance technology more effective in its role of monitoring, detecting and intercepting undesirable behaviour, but has also played a significant role in delivering information that can be used to enhance business solutions and optimise overall business performance. According to reports by IMS Research, growing demand for digital security has created a US\$ 40 billion industry.

Retail. Major demand for electronic security across the globe comes from retail, educational and public sectors. Retail is a major vertical market for the digital security and surveillance industry. Retail surveillance helps retail companies identify incidents of retail crimes like shoplifting, violence and vandalism. It is mostly carried out through CCTV cameras and has recently evolved to